Electromagnetism

Magnetic Flux Density

Magnetic flux density is the density of the magnetic field in an area normal to the field lines, with the units Tesla[T], while the actual flux itself has units Weber[Wb]

Magnetic flux can be calculated as so:

$$B[T] = \frac{\Phi[Wb]}{A[m^2]}$$

where B is the flux density, phi is the magnetic flux and A is the (effective) area

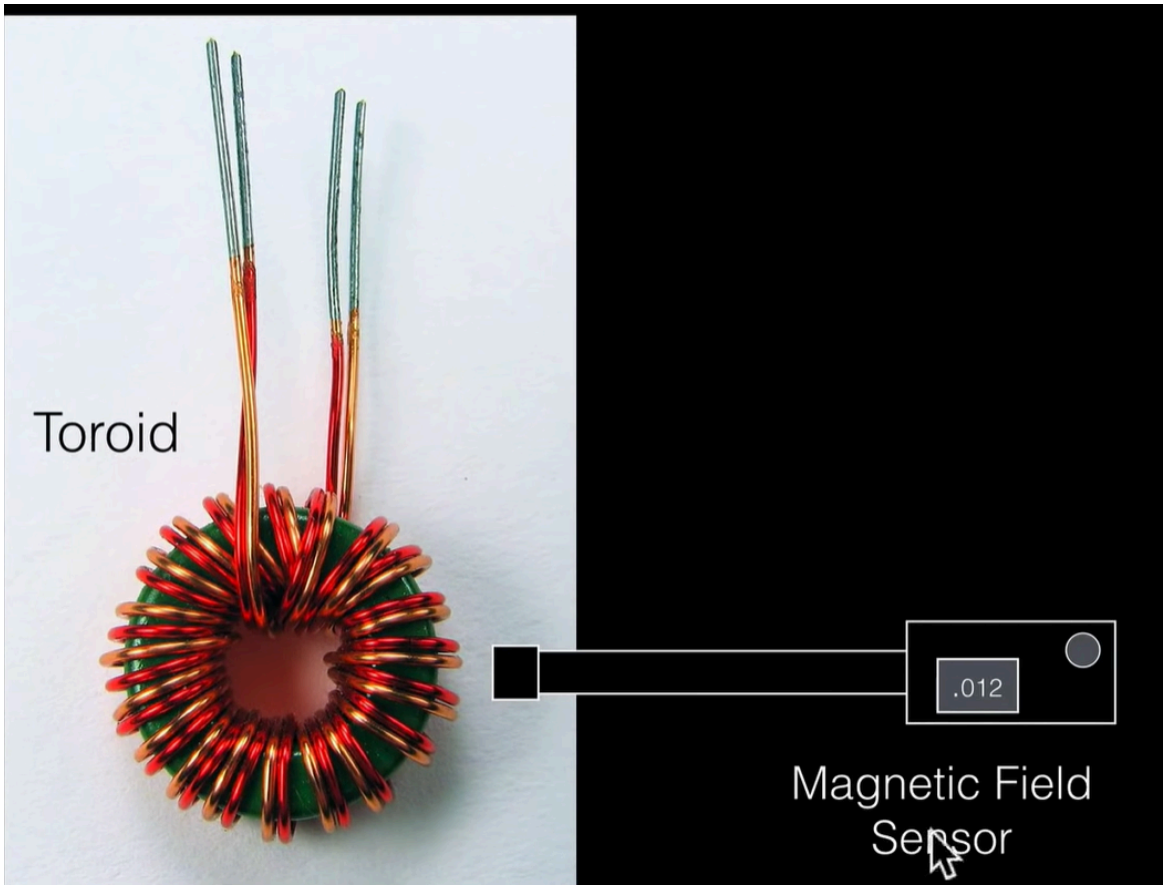
Magnetic flux (density) essentially measures the strength of a magnetic source, the higher the magnetic flux (density) is, the stronger the magnet is.

Permeability

Permeability refers to how easily a magnetic field can be established in the material, aka how easily the material can be made into an electromagnet.

Permeability has units Wb/(Atm), or magnetic flux per ampere (turn) meters

we can think of this unit as measuring the amount of magnetic flux established in a material when a fixed current flows through a solenoid wrapped around the material.



an alternative unit is H/m (henries/meter), as henries are defined as Wb/A.

For non magnetic materials, the magnetic flux measured will be very low and for ferromagnetic materials, the magnetic flux measured will be very high. Thus, instead of recording the permeability directly, we can measure permeability relative to a vacuum.

Permeability of a vacuum/air refers to same setup above but without any material in the coils, and is a constant of

$$\mu_0 = 4\pi \times 10^{-7} Wb/Atm$$

Using the constant, we can measure something called relative permeability, which is more useful for giving an idea of how easy something can be a electromagnet/ how magnetic something is:

$$\mu_r = \frac{\mu}{\mu_0}$$

Thus, we can expect for non magnetic materials like wood, its permeability and the permeability of a vacuum is about the same, so the relative

permeability is ~1

magnetic materials will above have a higher relative permeability. Shown are some permeability values of some materials:

Material	permeability	relative permeability
wood	1.25663760×10^-6	1.00000043
aluminium	1.256665×10^−6	1.000022
Nickel	(1.26 to 7.54)×10^−4	100 - 600
iron	2.5×10^-1	200000

Note how wood has a relative permeability of ~1, and non magnetic materials have a slightly higher relative permeability, but still ~1 and ferromagnetic materials have high relative permeabilities.

Reluctance

Reluctance is the opposition to establish a magnetic field in a material. This is similar to resistance, and has a similar formula.

$$\mathcal{R} = \frac{l}{\mu A}$$

Note how reluctance is directly proportional to length as the flux has to travel through more of the material, and that reluctance is inversely proportional to the area as a higher area means more flux can flow through the material, decreasing reluctance.

Reluctance has units ampere(turns) per weber, or At/Wb

Magnetomotive force

magnetomotive force (mmf) is the force that creates a magnetic field, similarly to voltage (electromotive force, emf)

For a solenoid,

$$\mathcal{F}_m = NI$$

where N is the number of coils and I is the current. mmf has units Atm (Ampere turn meters)

we can think of mmf, magnetic flux and reluctance to be analgous to voltage, current and resistance

Ohms law for electricity:

$$V = IR$$

Ohms law for magnetism:

$$\mathcal{F}_m = \Phi\mathcal{R}$$

we can think of mmf providing the magnetic flux, like how a battery would provide current, and magnetic flux flowing through the material like how current flows through a wire.

Strengthening electromagnets

For an electromagnet, we can find flux density as so:

$$B = \frac{\mu NI}{l}$$

where l is the length of the material. we can see that a higher permeability, more coils and higher current increase the flux density, while a longer length decreases flux density (as the magnetic flux is less concentrated throughout the material.)

Magnetising Force

magnetising force (H) is the magnetomotive force per unit length of a magnetic circuit, in other words

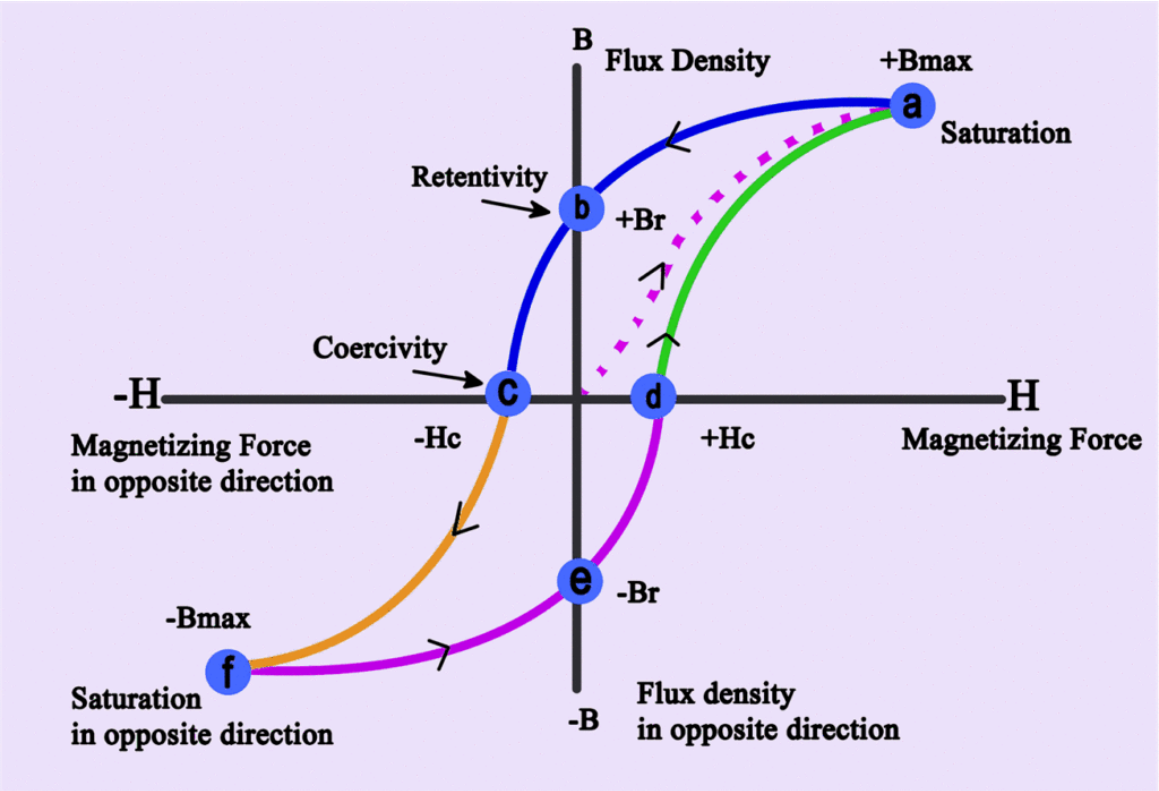
$$H = \frac{\mathcal{F}_m}{l} = \frac{NI}{l}$$

This also shows that H is only dependent on the number of turns, current and the length, and is not affected by the material itself.

fuck you

Magnetic Hysteresis & Hysteresis Curve

Hysteresis is when the magnetisation of a material (B) lags from the applied magnetic force(H), which mainly depends on the input current



The Hysteresis/B-H curve shows the relationship between the two values, where the positive/negative polarity of B represents a swap in magnetic polarity, while the positive/negative polarity of H represents the direction of input current and hence the direction of the magnetising force.

The curve represents a setup with a solenoid wrapped around an originally unmagnetised magnetic core, where the current can be varied. and the magnetic flux of the core is measured.

There are 7 phases of this diagram:

Phase 1: Origin to A

Initially, as the core is not magnetised and no current is applied, no magnetising force is applied and hence no magnetic flux is formed. (i.e. B=0,H=0 at origin)

As a current & magnetising force is applied, the flux density increases proportionally up to a point a, where it starts to level out. We call this saturation, as we cannot increase the flux density anymore even if we add more magnetising force. Thus, we also call point a the saturation point/maximum point

Phase 2: a to b

As H decreases, B will also decrease but will decrease at a slower rate and not fall back to 0Wb(which we refer to B_r) This means that even though we had removed the magnetising force, we "permanently" magnetised our core with the solenoid, as some of the domains are still aligned, however with lesser flux density than what we had put in.

This effect of staying magnetised even after an external magnetic force is removed is called retentivity (retention of flux density)

Phase 3: b to c

Now, we then change the direction of current, so that H is negative (opp direction)

This causes the flux density to decrease more to point c, aka when flux density is 0.

This can be thought of as the opposite magnetic force messing up the domains, causing them to no longer be aligned together, thus having B=0

The point where c intersects on the H axis is the coercive force, $-H_c$, which is the magnetising force required to remove the retentivity and make B=0.

Phase 4: c to f

If we further increase current and H in the opposite direction, the flux density will also increase similarly in phase 1, however in the opposite direction. B will then continue to increase in the negative direction until it reaches saturation again, at the same magnitude in phase 1. Similar properties of saturation apply here.

Phase 5: f to e

if we decrease H back to 0 again, B will also decrease at a slower rate, reaching $-B_r$.

Phase 6: e to d

increasing H back to the positive direction causes the flux density to reach 0, where the curve cuts at $+H_c$.

Phase 7: d to a

Similar to phase 1, as we increase H again however from a non 0 value, B will reach the saturation point.

Phases 2 to 7 will then repeat for when H changes.

Note that different materials will have differently shapes B-H curves

From this curve we now know of 3 properties of magnetisation, permeability, retentivity and coercivity

Permeability: How easily a magnetic field can be setup

Retentivity: How well a magnetic field can be maintained/retained (w/o an external field)

Coercivity: How hard it is to demagnetise a magnetic field.

high retentivity and coercivity is useful for permanent magnets, where you want to maintain the magnetic field.

High permeability and low coercivity is useful for electromagnetic alarms and circuit breakers, where a core needs to be rapidly magnetised and demagnetised.

High coercivity is also useful for credit cards, so that information is harder to erase, while low coercivity is useful for single trip train cards, where information can be easily erased to be reused again

Faraday's Law

When a solenoid experiences a changing magnetic flux, it will induce a voltage across the solenoid opposite to the changing magnetic flux. i.e.

$$V_{ind} = N\frac{d\Phi}{dt}$$

where N is the number of coils.

Lenz's Law

The induced voltage across a solenoid causes a current that creates a magnetic flux which opposes the changing magnetic field.

In summary, Faraday's Law talks about the magnitude while Lenz's law talks about the direction.

Quantity	Units
Magnetic Flux	Weber [Wb]

Magnetic Flux Density	Tesla [T]
Permeability	Wb/Atm
Reluctance	At/Wb
Magnetomotive force	Ampere-turn meters[Atm]
Magnetising Force	Ampere-turns [At]